

Distinct Activation Mechanisms of CXCR4 and ACKR3 Revealed by Single-Molecule Analysis of their Conformational Landscapes

Reviewed Preprint

v1 • August 21, 2024

Not revised

Christopher T Schafer, Raymond F Pauszek III, Martin Gustavsson, Tracy M Handel , David P Millar 

Skaggs School of Pharmacy and Pharmaceutical Sciences, Department of Pharmacology, University of California San Diego, La Jolla, Unites States • Department of Integrative Structural and Computational Biology, The Scripps Research Institute, La Jolla, Unites States

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

Abstract

The canonical chemokine receptor CXCR4 and atypical receptor ACKR3 both respond to CXCL12 but induce different effector responses to regulate cell migration. While CXCR4 couples to G proteins and directly promotes cell migration, ACKR3 is G protein- independent and scavenges CXCL12 to regulate extracellular chemokine levels and maintain CXCR4 responsiveness, thereby indirectly influencing migration. The receptors also have distinct activation requirements. CXCR4 only responds to wild-type CXCL12 and is sensitive to mutation of the chemokine. By contrast, ACKR3 recruits GPCR kinases (GRKs) and β -arrestins and promiscuously responds to CXCL12, CXCL12 variants, other peptides and proteins, and is relatively insensitive to mutation. To investigate the role of conformational dynamics in the distinct pharmacological behaviors of CXCR4 and ACKR3, we employed single-molecule FRET to track discrete conformational states of the receptors in real-time. The data revealed that apo-CXCR4 preferentially populates a high- FRET inactive state, while apo-ACKR3 shows little conformational preference and high transition probabilities among multiple inactive, intermediate and active conformations, consistent with its propensity for activation. Multiple active-like ACKR3 conformations are populated in response to agonists, compared to the single CXCR4 active-state. This and the markedly different conformational landscapes of the receptors suggest that activation of ACKR3 may be achieved by a broader distribution of conformational states than CXCR4. Much of the conformational heterogeneity of ACKR3 is linked to a single residue that differs between ACKR3 and CXCR4. The dynamic properties of ACKR3 may underly its inability to form productive interactions with G proteins that would drive canonical GPCR signaling.

eLife assessment

This manuscript describes the characterization of the conformational dynamics of two chemokine receptors at the single-molecule level using FRET. The authors make a **convincing** case for attributing the distinct interaction and pharmacology of the two receptors to differences in their conformational energy landscape. These **important** findings will be of interest to scientists working on activation mechanisms of GPCRs and signal transduction.

<https://doi.org/10.7554/eLife.100098.1.sa3>

Introduction

CXC chemokine receptor 4 (CXCR4) is one of the most intensively studied chemokine receptors due to its central role in driving cell migration during development and immune responses, and in cancer where it promotes tumor growth and metastasis (1–4). As a class A G protein-coupled receptor (GPCR), CXCR4 activates inhibitory Gai protein signaling pathways to directly control cell movement in response to the chemokine CXCL12 (5). The activated receptor is also phosphorylated by GPCR kinases (GRKs), which promotes arrestin recruitment and cessation of the G protein signal. CXCR4 often works together with atypical chemokine receptor 3 (ACKR3, formerly CXCR7) which indirectly influences migration by scavenging CXCL12 to regulate available extracellular levels of the agonist and in turn the responsiveness of CXCR4 (6–8). In the absence of ACKR3 scavenging, excessive CXCL12 stimulation of CXCR4 leads to downregulation, resulting in profound effects on neuronal cell migration and development (9–11).

In contrast to CXCR4 and with some exceptions noted (12, 13), ACKR3 lacks G protein activity and instead is considered to be arrestin-biased (14). However, we and others (9) have shown that arrestins are dispensable for chemokine scavenging while GRK phosphorylation is critical for this function (15), suggesting that ACKR3 might be better described as GRK-biased. The molecular basis for its inability to couple to G proteins in most cell types remains an unanswered question. Our recently determined structures of ACKR3-ligand complexes showed the expected hallmarks of GPCR activation including “microswitch residues” in active state configurations, displacement of transmembrane helix 6 (TM6) away from the helical bundle and an open intracellular pocket, consistent with a receptor that should be able to activate G proteins (16). Accordingly, we replaced the intracellular loops (ICLs) of ACKR3 with those of CXCR2, a canonical G protein-coupled chemokine receptor; however, these changes did not lead to G protein activation (16), suggesting that the lack of coupling is not due to the absence of specific residue interactions. CXCL12 also adopts a distinct pose when bound to ACKR3 compared to CXCR4 and all other chemokines in chemokine-receptor complexes, but since small molecules induce similar biased effector responses, the chemokine pose cannot explain the effector-coupling bias (16). Having excluded other mechanisms we therefore surmised that the inability of ACKR3 to activate G proteins may be due to differences in receptor dynamics. Consistent with this hypothesis, the ICLs observed in ACKR3-agonist complexes are disordered, which may preclude productive effector coupling. The dynamic nature of ACKR3 is also suggested by its considerable constitutive activity in recruiting β -arrestins and its high level of constitutive internalization (16–18).

In addition to their distinct effector interactions, ACKR3 and CXCR4 have dramatically different susceptibilities to activation by different ligands. CXCR4 is activated by a single chemokine agonist, CXCL12. Moreover, modifications of the CXCL12 N-terminal signaling domain (e.g. single point mutations as in CXCL12P2G or multiple mutations as in CXCL12LRHQ) transform the chemokine

into an antagonist (19, 20). Mutational analysis, modeling, and new structures of CXCR4 suggest that ligand activation involves a precise network of interacting residues that stabilize the active receptor conformation (21–25). By contrast, ACKR3 is activated by CXCL12 as well as its variants (including CXCL12P2G and CXCL12LRHQ (19, 20)), other chemokines (CXCL11) (26), other proteins (adrenomedullin, BAM22) (27, 28), and opioid peptides (29). In fact, most ligands for ACKR3 act as agonists, which is best explained by a non-specific “distortion” mechanism of activation whereby any ligand that breaches the binding pocket causes helical movements that are permissive to GRK phosphorylation and arrestin recruitment. A distortion mechanism is consistent with the different binding poses observed for a small molecule agonist compared to the CXCL12 N-terminus in the receptor orthosteric pocket (16), and ligand bulk rather than specific interactions between agonist and ACKR3 being required for activation. We hypothesize that this distortion mechanism would also be facilitated by a receptor that is conformationally dynamic by enabling nonspecific ligand interactions and more than a single conformation to promote activation.

To investigate the role of conformational dynamics in the distinct pharmacological behaviors of ACKR3 and CXCR4, we developed a single-molecule Förster resonance energy transfer (smFRET) approach. Many ensemble methods such as EPR, NMR and fluorescence-based methods have provided considerable insights into GPCR dynamics, conformational heterogeneity, and exchange between distinct structural states (30–36). However, these methods are limited in their ability to resolve the sequence of state-to-state transitions except in rare cases where transitions can be temporally coordinated with high precision (37). By contrast smFRET enables detection of sparsely populated states, reveals the sequence of state-to-state transitions and provides kinetic information through analysis of state dwell times. For example, smFRET studies of the β 2 adrenergic receptor (β 2AR) revealed a dynamic equilibrium between inactive and active conformations that was responsive to agonist and G protein binding (38). More recent smFRET studies of the glucagon receptor (39), the A2A receptor (A2AR) (40, 41), and the μ -opioid receptor (42) have documented the existence of stable intermediate conformations in addition to inactive and active receptor conformations.

Here we present the first smFRET study of the chemokine receptors CXCR4 and ACKR3. Our experimental system allows real-time observation of the conformational fluctuations of individual receptor molecules in a native-like lipid environment and assessment of the differences in the conformational dynamics of the two receptors in their apo states and in response to ligands. Our results indicate that ACKR3 is more dynamic and conformationally heterogeneous than CXCR4 or other class A GPCRs previously studied, which may explain its activation prone nature and lack of G protein coupling. In contrast, CXCR4 appears less flexible, consistent with a more restricted, structurally-defined activation mechanism. Together these data characterize the molecular differences between CXCR4 and ACKR3 and suggest that enhanced conformational dynamics plays an important role in the atypical function of ACKR3.

Results

Development of smFRET experimental systems for CXCR4 and ACKR3

To visualize conformational fluctuations of CXCR4 and ACKR3 by smFRET, cysteine residues were introduced into CXCR4 at positions 150^{4.40} in TM4 and 233^{6.29} in TM6, and at 159^{4.40} and 245^{6.28} in ACKR3 (Figs. S1A & B) (numbers in superscript refer to the Ballesteros-Weinstein numbering scheme for GPCRs) for covalent labeling with FRET donor (D, Alexa Fluor 555, A555) and acceptor (A, Cyanine5, Cy5) fluorophores. Single receptor molecules of labeled CXCR4 or ACKR3 were reconstituted into phospholipid nanodiscs to mimic the native membrane bilayer environment. Nanodiscs have been used in a wide variety of structural and biophysical studies of integral

membrane proteins, including ACKR3, and reconstitute protein-lipid interactions lost in detergent systems (43–45). The nanodisc-receptor complexes were then tethered to a quartz slide through biotinylation of the nanodisc membrane scaffolding protein (MSP) (Fig. 1A). To promote monomeric receptor incorporation in each nanodisc, we utilized MSP1E3D1, which forms nanodiscs with a diameter of approximately 13 nm (46, 47).

Crystal and cryo-electron microscopy (cryo-EM) structures of homologous class A GPCRs, such as β 2AR, in inactive and active conformations, reveal that TM6 moves outwards from the TM helical bundle during activation, whereas the position of TM4 remains relatively fixed (48). Accordingly, we anticipated that labeling at the positions indicated above would be sensitive to transitions between inactive and active receptor conformations and give rise to different donor-acceptor distances and FRET efficiencies that could be resolved by smFRET measurements (shown schematically in Fig. 1B). These positions are similar to those used effectively for β 2AR (38), A2AR, (40), and the μ -opioid receptor (42) for monitoring their conformational dynamics by smFRET. Importantly, neither wild-type (WT) CXCR4 nor ACKR3 exhibited significant labeling and the double cysteine receptor mutants retained the ability to recruit β -arrestin2 (Figs. S1C & D).

ACKR3 exhibits greater conformational dynamics than CXCR4

Receptor-nanodisc complexes were imaged on the slide surface using smFRET microscopy by exciting the A555 donor with a green (532 nm) laser and monitoring the resulting emission from both A555 and the Cy5 acceptor over time on separate segments of a CCD camera. Several hundred individual receptor-nanodisc complexes were typically observed in the field of view. Individual receptor molecules labeled with a single A555 donor and a single Cy5 acceptor were identified by single-step photobleaching transitions (abrupt loss of acceptor fluorescence signal or simultaneous loss of fluorescence signal in both channels), as shown for two representative ACKR3-nanodisc complexes in Fig. 1C. Additionally, anti-correlated changes in donor and acceptor emission prior to photobleaching confirmed that FRET occurred between A555 and Cy5. Additional examples of single-molecule traces for ACKR3 and CXCR4 are shown in Fig. S2.

FRET data from many individual receptor-nanodisc complexes, recorded in the presence or absence of different ligands, were globally analyzed using Hidden Markov Models (HMMs) assuming the presence of two, three, four, or five FRET states (Materials and Methods). To evaluate the appropriate level of model complexity, each resulting FRET efficiency distribution was fit with a Gaussian Mixture Model (GMM) and the corresponding Bayesian Information Criterion (BIC) was calculated. The BIC is a statistical measure of the likelihood that the model describes the data while also penalizing the addition of parameters that could lead to overfitting of noise. Theoretically, this value will be at a minimum for the model with the appropriate number of states. As an additional criterion, we carefully compared the FRET distributions recovered for the different models, looking for consistent peak positions across different experimental conditions.

These analyses indicated that three FRET states were sufficient to fit the smFRET data for CXCR4 under all conditions (Figs. S3 & S4). The resulting FRET efficiency distributions are presented in Figs. 2A & B. The predominant high-FRET state ($E = 0.85$) observed in the apo-state (Fig. 2A) suggests that TM4 and 6 are in close physical proximity, which is consistent with the conformation of inactive GPCRs (49, 50). Accordingly, we interpret this FRET state as the inactive conformation of CXCR4 and designate it as R. The minor low-FRET state ($E = 0.19$) reflects outward movement of TM6 away from TM4, as expected for an active receptor conformation (designated R*) (48, 51) and consistent with limited basal activity of CXCR4. Moreover, there was a major population shift from the high-FRET state to the low-FRET state upon addition of CXCL12WT (Fig. 2B), consistent with the expected stabilization of active GPCR conformations by agonists (52, 53). To gain insight into the nature of the mid-FRET state ($E = 0.59$), we examined the connectivity between all three FRET states. Two-dimensional transition density probability (TDP) plots revealed that the three FRET states were connected in a sequential fashion (Figs. 2A & B). Notably, non-sequential transitions that “skip over” an intervening FRET state were rarely

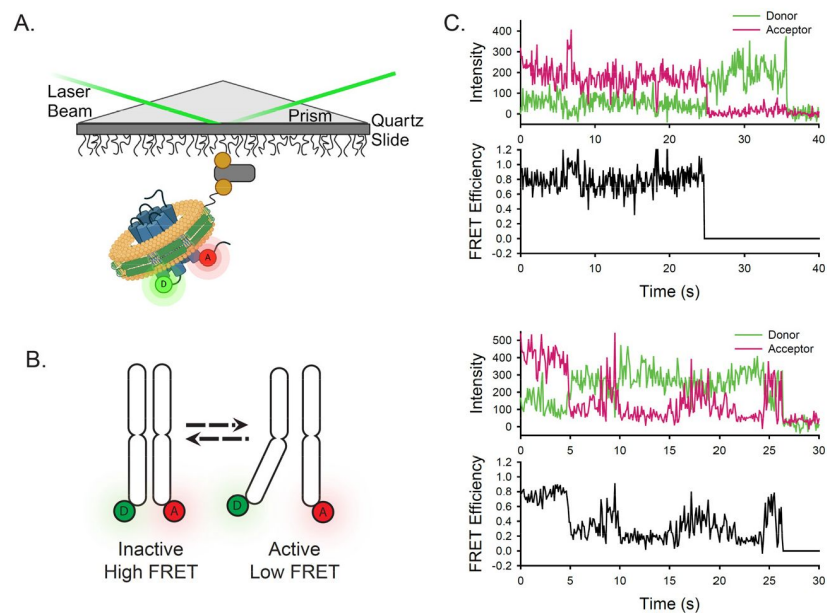


Figure 1.

Experimental design of the smFRET system. A) A single receptor molecule (blue) was labeled with donor (D) and acceptor (A) fluorophores, inserted into a phospholipid (yellow) nanodisc (green), and immobilized on a quartz slide via biotin (brown circle)-neutravidin (grey rectangle) attachment. A prism facilitates total internal reflection of the excitation laser to excite only donor fluorophores close to the surface. B) Cartoon depicting inactive (left) and active (right) receptor conformations. C) Two representative single-molecule time traces for apo-ACKR3. In both examples, the donor (green) and acceptor (red) intensities are shown in the top panel and the corresponding FRET efficiency (black) is shown in the bottom panel.

observed. These results suggest that the mid-FRET state represents an intermediate receptor conformation (designated R') that lies on the pathway between inactive and active conformations. In the apo-state, transitions were mostly observed between states R and R' (Fig. 2A), while in the presence of CXCL12WT the most frequent transitions were observed between the R' and R* states (Fig. 2B).

In contrast, the FRET efficiency histogram for ACKR3 in the apo state (Fig. 2C) was much broader than the corresponding histogram for CXCR4 (Fig. 2A), indicating greater conformational heterogeneity. Moreover, the smFRET data for ACKR3 could not be described by three FRET states and the inclusion of a fourth state was necessary (Figs. S5 & S6). The FRET distributions recovered by four-state global analysis are presented in Figs. 2C & D. The positions of the high-FRET ($E = 0.85$) and low-FRET ($E = 0.11$) peaks are similar to those observed in CXCR4 and are likewise assigned to inactive (R) and active (R*) conformations, respectively. Consistent with these assignments, the R* state increased in population at the expense of the R state in the presence of the chemokine agonists CXCL12WT (Fig. 2D) or CXCL11 (Fig. S7B). TDP plots indicated that the four FRET states in ACKR3 were connected in a sequential fashion (Figs. 2C & D). Accordingly, the mid-FRET peaks ($E = 0.39$ and $E = 0.66$) are assigned to two intermediate receptor conformations, designated R'' and R', respectively. Notably, the R'' state was not observed in CXCR4 (Figs. 2A & B). The R' state in ACKR3 decreased in population in the presence of chemokine agonists (Fig. 2D, Fig. S7B), suggesting that this state represents an inactive intermediate receptor conformation, consistent with its position on the conformational pathway (closest to R). In striking contrast to apo-CXCR4, apo-ACKR3 populated the four conformational states more or less equally and all possible sequential conformational transitions were observed (Fig. 2C). Thus, ACKR3 is intrinsically more conformationally heterogeneous and dynamic than CXCR4. In the presence of CXCL12WT, ACKR3 showed more frequent R' $\leftrightarrow$ R'' and R'' $\leftrightarrow$ R* transitions compared with the apo-receptor, accounting for the population shift towards the R* state (Figs. 2C & D).

Effect of small-molecule ligands on conformational states of CXCR4 and ACKR3

The small-molecule ligand IT1t is reported to act as an inverse agonist of CXCR4 (51–53). However, the conformational distribution of CXCR4 showed little change with inclusion of IT1t (Figs. 3A & B, Fig. S8), suggesting that it acts as a neutral antagonist in our purified in vitro system. However, R $\leftrightarrow$ R' transitions were more probable in the presence of IT1t compared with the apo-receptor (Figs. 3A & B).

Treatment of ACKR3 with the small-molecule agonist VUF15485 (53) shifted the conformational distribution of the receptor towards the active R* state (Figs. S7C & S8), as expected for an agonist and supporting the assignment of the R* FRET state. In contrast, treatment of ACKR3 with the small-molecule inverse agonist VUF16840 (53) shifted the conformational distribution to the inactive R conformation, with a concomitant decrease of R* (Fig. 3D, Fig. S8), consistent with the suppression of the basal activity of ACKR3 (53). The R'' population also decreased in the presence of the inverse agonist (Fig. 3D, Fig. S8), suggesting that this state represents an active-like receptor conformation, consistent with its placement on the conformational pathway (closest to R*). The suppression of active receptor conformations was also evident in the TDP, which revealed fewer transitions between R'' and R* conformations relative to the apo receptor (Figs. 3C & D).

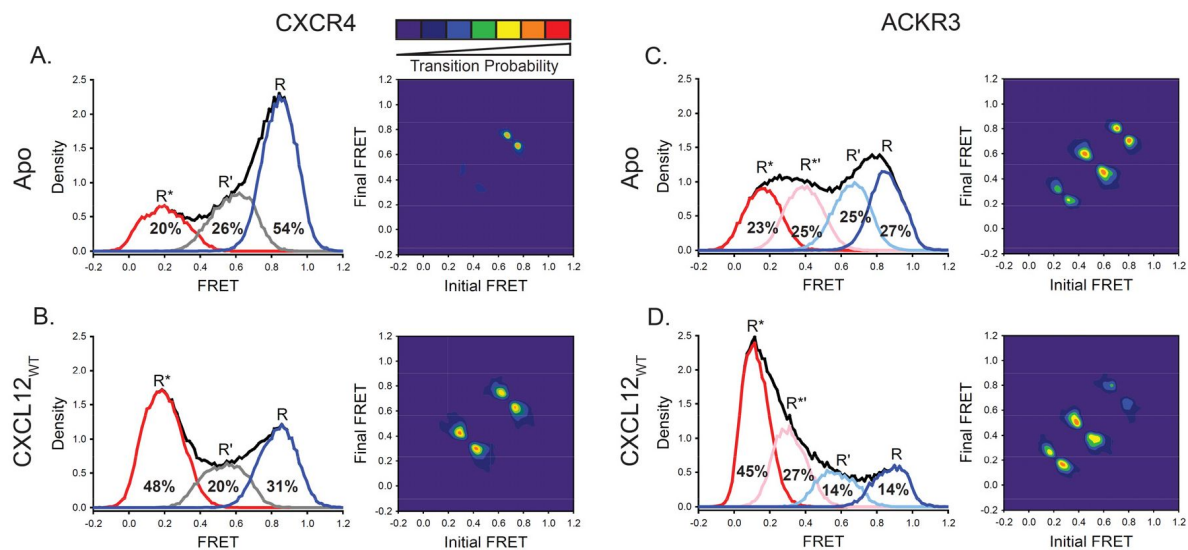


Figure 2.

ACKR3 exhibits greater conformational flexibility compared to CXCR4. A) FRET efficiency histogram of apo-CXCR4 (left, black trace) resolved into three distinct conformational states: a high-FRET state corresponding to the inactive receptor conformation (R , blue), a low-FRET active receptor conformation (R^* , red) and an intermediate conformation (R' , gray). The fractional populations of each state obtained from global analysis are indicated. The receptor is mostly in the inactive conformation. A transition density probability (TDP) plot (right) displays the relative probabilities of transitions from an initial FRET state (x-axis) to a final FRET state (y-axis). For apo- CXCR4, transitions between R and R' states are observed most frequently. B) Addition of CXCL12 to CXCR4 shifted the conformational distribution to the low-FRET R^* state and resulted in more transitions between all three FRET states. C) The broad FRET efficiency histogram of apo-ACKR3 (left, black trace) is resolved into four distinct conformational states: inactive R (blue), active R^* (red), inactive-like R' (light blue), and active-like R'' (pink). Little conformational preference is observed among these states. Moreover, all possible sequential state-to-state transitions are observed (right). D) Addition of CXCL12 to ACKR3 shifted the conformational distribution to the low-FRET R^* state, which was also reflected in the transition probabilities. In all cases, data sets represent the analysis of at least three independent experiments.

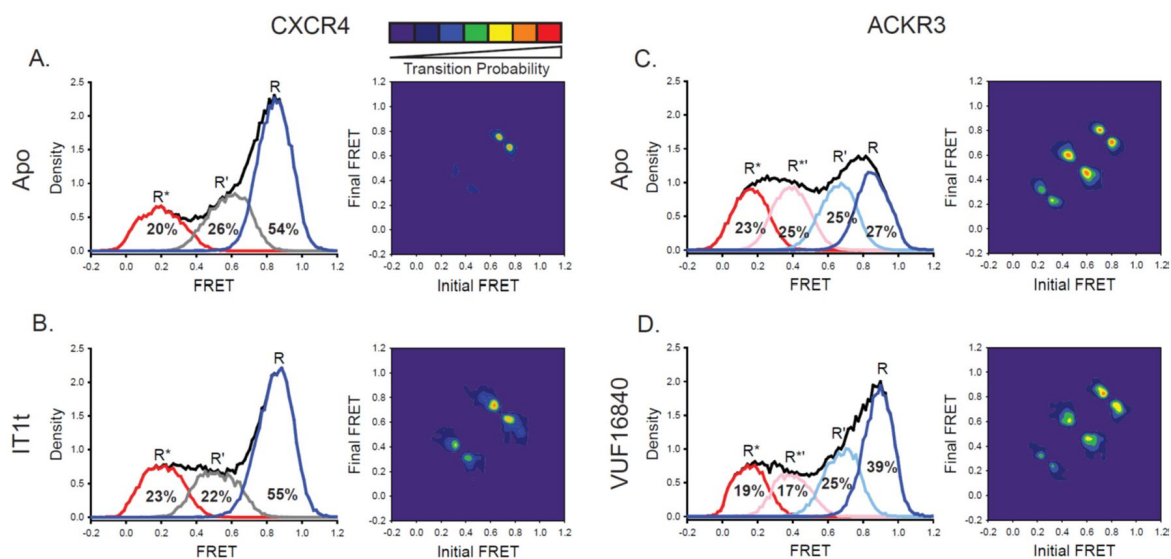


Figure 3.

A small molecule inhibitor shifts the ACKR3 conformational population to the inactive FRET state, while CXCR4 is largely unaffected. A) FRET distributions and TDP of apo-CXCR4 repeated from [Fig. 2A](#) for comparison. B) Treatment of CXCR4 with the inhibitor IT1t had little impact on the FRET distribution (compared to 2A), but increased transition probabilities compared to the apo-receptor. C) FRET distributions and TDP of apo-ACKR3 repeated from [Fig. 2C](#) for comparison. D) Treatment of ACKR3 with VUF16840, an inverse agonist, shifted the conformational distribution and TDPs to the high-FRET inactive R conformation. Data sets represent the analysis of at least three independent experiments. The overall FRET efficiency envelopes for the samples are represented by the black traces.

CXCL12 N-terminal mutants promote active receptor conformations despite their contrasting pharmacological effects on CXCR4 and ACKR3

CXCR4 is sensitive to N-terminal mutations of CXCL12 while ACKR3 is relatively insensitive. For example, the variant CXCL12P2G, containing a proline to glycine mutation in the second position, and CXCL12LRHQ, where the first three residues of CXCL12WT are replaced with the four-residue motif LRHQ starting with L0, are antagonists of CXCR4 but agonists of ACKR3 (19, 20). To gain further insight into the ligand-dependent responses of CXCR4 and ACKR3, we examined how these mutant chemokines influence the conformational states and dynamics of both receptors.

Surprisingly, despite acting as a CXCR4 antagonist (20), CXCL12P2G promoted a shift to the active R* conformation of CXCR4 compared to the apo-receptor (R* increased by 16%, **Figs. 4A & B**, Fig. S8), although the shift was less pronounced than observed for CXCL12WT (28% increase in R*, **Fig. 2B**, Fig. S8). The shift is also evident in the TDPs where the state-to-state transitions involving the R* active state were more probable for the CXCL12P2G complex compared with apo-CXCR4 (**Figs. 4A & B**). The presence of CXCL12LRHQ had a more subtle effect on CXCR4: the active R* conformation increased by 9% (**Figs. 4A & C**, Fig. S8). Despite the ability of CXCL12P2G and CXCL12LRHQ to stabilize the active R* conformation of CXCR4, both variants are known to act as antagonists (20). The implication is that outward movement of TM6 and consequent opening of the intracellular cleft is not sufficient to drive G protein activation of CXCR4. According to a previously proposed model, a precise network of interacting residues within both CXCL12 and CXCR4 is required for receptor activation and signaling (22). Our results reveal that disrupting this network with CXCL12 mutations P2G and LRHQ still promoted the active receptor conformation, suggesting that conformational changes leading to activation can be separated from formation of the interaction network.

In the case of ACKR3, CXCL12P2G induced a modest increase in the population of the active R* conformation relative to the apo-receptor (R* increased by 5%, **Figs. 4D & E**, Fig. S8), consistent with the ability of this CXCL12 variant to act as an agonist of ACKR3 (20). Additionally, CXCL12P2G also promoted formation of the active-like intermediate R** conformation (R** increased by 6%, **Figs. 4D & E**, Fig. S8), suggesting that activation of ACKR3 can be achieved by populating the R** state, not just the R* state, which is consistent with a flexible, distortion activation mechanism. Together, CXCL12P2G increased active (R*) and active-like (R**) conformations by 11 %, somewhat less than observed for CXCL12WT (24%, **Fig. 2C**, Fig. S8). CXCL12LRHQ also promoted the active (R*) and active-like (R**) conformations of ACKR3 (R* + R** increased by 11%, **Fig. 4F**, Fig. S8). Additionally, transitions to the active R* and active-like R** conformations appeared to be more probable in the presence of CXCL12LRHQ compared to CXCL12P2G (**Figs. 4E & F**), which could be a consequence of its longer residence time on the receptor (54). The agonism observed for both chemokine variants (19, 20) suggests that both the active R* and active-like R** conformations of ACKR3 are sufficient for GRK phosphorylation and arrestin recruitment.

ACKR3 constitutive activity is linked to receptor conformational heterogeneity

As noted above, apo-ACKR3 displays little conformational selectivity, with similar occupancies observed for all four FRET states (**Fig. 2C**). We hypothesized that this might be related to the constitutive activity of the receptor (13, 16) and tied to the presence of Tyr at position 257^{6.40} (**Fig. 5A**), which is a hydrophobic residue (V, I, or L) in all other chemokine receptors. In many other class A GPCRs, mutating the residue at 6.40 results in constitutive activity (32, 55–57) and previous analysis of rhodopsin suggests that this is a consequence of lowering the energy barrier for transitions between different receptor conformations (58). In our previous work, we

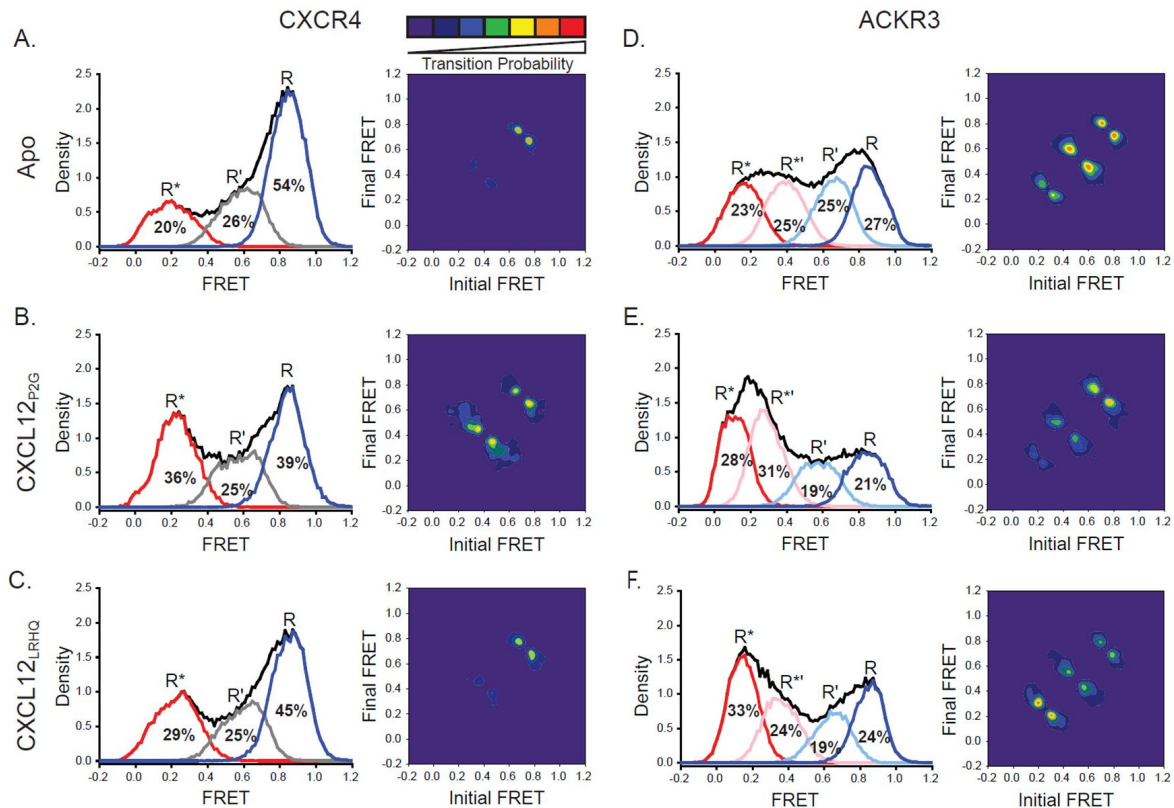


Figure 4.

CXCL12 variants containing mutations to the N-terminus promoted active receptor conformations in both CXCR4 and ACKR3. A) FRET distributions and TDP for apo-CXCR4 repeated from **Fig. 2A** for reference. B) Addition of CXCL12P2G to CXCR4 promoted a shift to the low-FRET active (R*) conformation and an increase in state-to-state transition probabilities. C) CXCL12LRHQ led to a more subtle shift to the R* conformation of CXCR4 without affecting the transition probabilities. D) FRET distributions and TDPs for apo-ACKR3 repeated from **Fig. 2C**. E) Treatment of ACKR3 with CXCL12P2G displayed a shift to low-FRET, R* and R'' states, while reducing the transition probabilities for R' $\leftrightarrow$ R'' and R'' $\leftrightarrow$ R* transitions relative to the apo-receptor. F) CXCL12LRHQ treatment of ACKR3 shifted the FRET distribution to the low-FRET R* active state and promoted R'' $\leftrightarrow$ R* transitions relative to the apo-receptor. In all cases, the data sets represent the analysis of at least three independent experiments. The overall FRET efficiency envelopes for the samples are represented by the black traces.

showed that mutation of Y257^{6.40} to leucine, the corresponding amino acid in CXCR4, reduces constitutive arrestin recruitment to ACKR3, while preserving the ability of the receptor to be activated by CXCL12 (Fig. S9) (16). Adding this single-point mutation to our ACKR3 smFRET construct had a significant impact, converting the broad conformational distribution of the WT apo-receptor to a narrower distribution concentrated in the high-FRET region (Figs. 5B & C). Overall, the FRET histogram is similar to what we observe for apo-CXCR4 (Fig. 2A), although the active conformation is still split between R* and R** states (the latter unique to ACKR3) (Fig. 5C). State-to-state transitions were also suppressed relative to WT ACKR3 (Figs. 5B & C). This result indicates that Y257^{6.40} is a major determinant of the broad conformational heterogeneity of ACKR3.

CXCL12WT promoted active-like conformations of Y257^{6.40}L ACKR3 (R* + R** increased by 16%, Figs. 5C & E), although the effect was somewhat reduced in comparison to WT ACKR3 (R* + R** increased by 24%, Figs. 5B & D). Consistent with this, the mutant receptor recruits β -arrestin in response to CXCL12WT with an E_{max} value that is only slightly reduced relative to the WT receptor (Fig. S9), suggesting again that the population of the R** intermediate conformational state may contribute to receptor activation.

Discussion

The molecular basis for the atypical pharmacological behavior of ACKR3 is not well understood. Since structures of ACKR3 show intracellular loop disorder, and progressive structural substitutions within the loops fail to promote G protein coupling (16), we recently proposed that the atypical nature of ACKR3 may be related to receptor conformational dynamics (16, 59). Consistent with this hypothesis, in the present study we found that the conformational dynamics of ACKR3 and the canonical GPCR CXCR4 are indeed markedly different. Our smFRET results revealed four distinct conformations of apo-ACKR3 with approximately equal populations: inactive R, active R*, R' (inactive-like) and R** (active-like). The state-to-state transition density probability plots (Fig. 2) further reinforced the notion that ACKR3 is a flexible receptor that readily exchanges between different conformational states, consistent with our previous structural studies where a flexible intracellular interface in the absence of interaction partners was observed (16). These observations imply that ACKR3 must have a relatively flat energy landscape, with similar local minima and low energy barriers between states (Fig. 6). The flat energy landscape is consistent with a dynamic receptor that is constitutively active and readily activated. By contrast, our smFRET results reveal that apo-CXCR4 is more conformationally static than ACKR3 and primarily populates an inactive high-FRET conformation with limited transitions and only a single intermediate. Consequently, CXCR4 must have a more rugged energy landscape with deeper valleys and higher barriers between states (Fig. 6).

The active-like R** conformational state appears to be unique to ACKR3. This state responds to agonist and inverse agonist ligands in the same manner as the active R* state. In contrast, the single intermediate state in CXCR4 (R') is not responsive to ligands. The presence of both active and active-like conformations in ACKR3 may be responsible for its activation prone nature and ligand promiscuity.

Structural features that make ACKR3 conformationally flexible include Y257^{6.40}, which we previously showed contributes to the constitutive activity of ACKR3 (16). Similar to the constitutively active M257^{6.40}Y mutant of rhodopsin, Y^{6.40} in ACKR3 stacks against Y^{5.58} and Y^{7.53} and stabilizes the active-like conformation (60). Additionally, the broad conformational distribution and high probability of state-to-state transitions of ACKR3 parallel the dramatically lower energy barrier of the M257^{6.40}Y rhodopsin mutant relative to WT rhodopsin (58). By contrast, mutation of Y257^{6.40} to leucine, the corresponding amino acid in CXCR4, promoted the

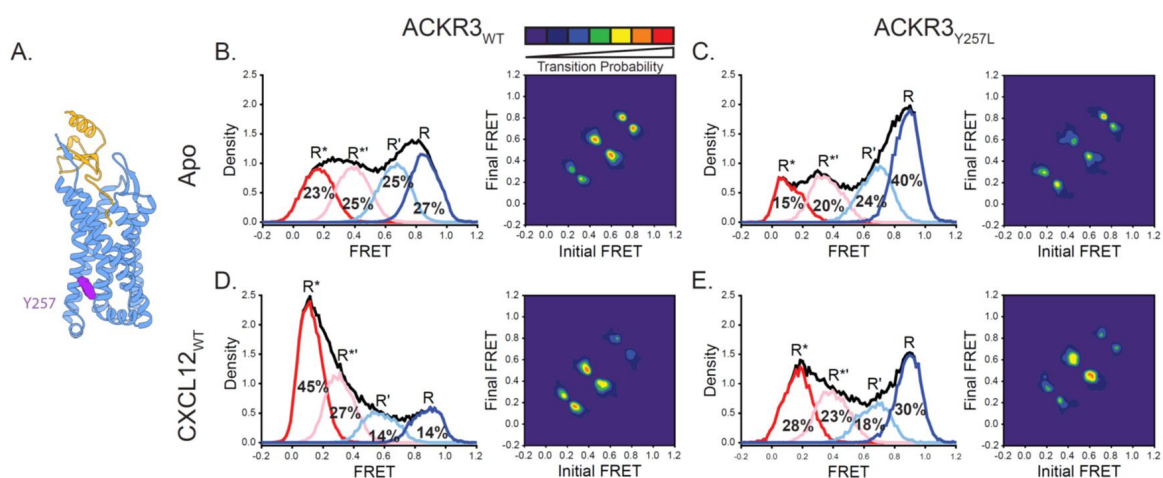


Figure 5.

Replacement of Y257^{6,40} with the corresponding residue in CXCR4 (leucine) reduces conformational heterogeneity of ACKR3. A) Structure of ACKR3 bound with CXCL12WT (PDBID: 7SK3) highlighting the location of Y257^{6,40} (purple) (16). B) FRET efficiency distributions and TDP of WT ACKR3 in the apo-state, repeated from Fig. 2C. C) The mutation Y257^{6,40}L shifted the conformational landscape of the apo-receptor to the high-FRET inactive R conformation at the expense of active R* and active-like R** conformations, and also reduced the probability of state-to-state transitions. D) FRET efficiency distributions and TDP of WT ACKR3 treated with CXCL12, repeated from Fig. 2D. E) Treatment of Y257^{6,40}L ACKR3 with CXCL12 promoted more low-FRET active R* and active-like R** states. Data sets represent the analysis of at least three independent experiments. The overall FRET efficiency envelopes for the samples are represented by the black traces.

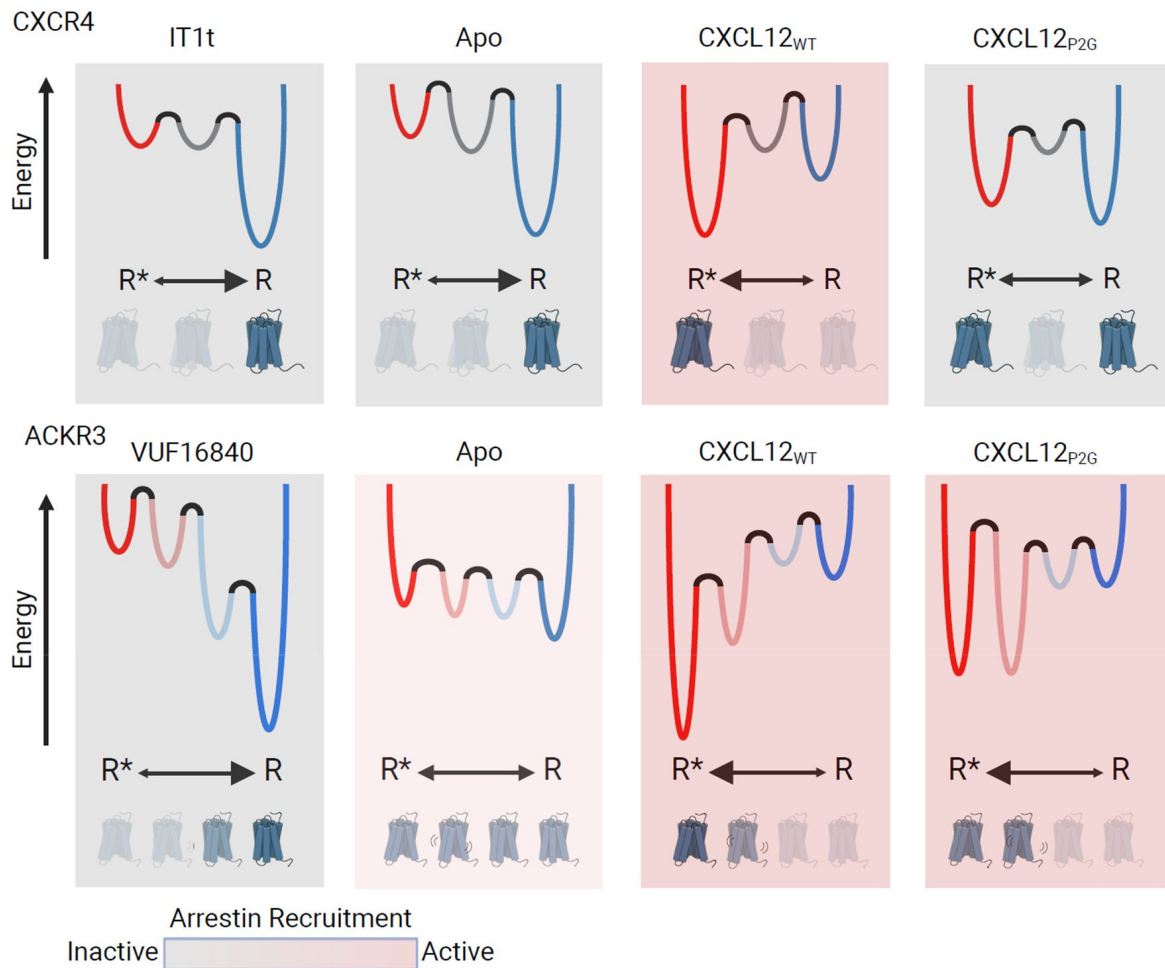


Figure 6.

Schematic of the conformational energy landscapes for CXCR4 and ACKR3, highlighting the differences in the responsiveness of the two receptors to ligands. CXCR4 populates three distinct conformations, shown here as wells on the energy landscape. Apo-CXCR4 is predominantly in the inactive R state. The receptor is converted incompletely to R* with CXCL12_{WT} treatment, while the antagonist IT1t has little impact on the conformational distribution. Though CXCL12_{P2G} is an antagonist, the ligand promoted a detectable shift to the active R* state, suggesting TM6 movement is not sufficient for CXCR4 activation. In contrast, ACKR3 populates four distinct conformations and shows little preference among them in the apo-form. The inverse agonist, VUF16840, shifts the population to the inactive R conformation, while the agonists CXCL12_{WT} and CXCL12_{P2G} promote the R* and R*' populations of ACKR3. Despite stabilizing different levels of the active R* state and active-like intermediate R*' state, both CXCL12_{WT} and CXCL12_{P2G} are agonists of ACKR3. The flexibility of ACKR3 may contribute to the ligand-promiscuity of this atypical receptor. Figure created with biorender.com

© 2024, BioRender Inc. Any parts of this image created with *BioRender* are not made available under the same license as the Reviewed Preprint, and are © 2024, BioRender Inc.

dominance of an inactive high-FRET receptor conformation similar to CXCR4 (Figs. 2A & 5C); it also reduced the transition probabilities (Figs. 5B & C) and the ability of ACKR3 to constitutively recruit arrestin (16).

In addition to the contribution of Y257^{6,40} to the dynamic behavior of ACKR3, a disulfide bond between C34 in the receptor N-terminus and C287 in extracellular loop 3 (ECL3), observed in all other reported chemokine receptor-chemokine structures (61–63) is conspicuously missing in cryo-EM structures of ACKR3 with chemokine and small molecule agonists (16). Furthermore, ACKR3 remains functional in the absence of these cysteines (64) unlike other chemokine receptors (65, 66). The disulfide may constrain the relative positions of TM1 and TM6/7 and the opening of the orthosteric pocket; therefore, its absence may confer ACKR3 with greater conformational flexibility than other chemokine receptors, consistent with our observations. It may also allow ACKR3 to be activated by diverse ligands.

The propensity of ACKR3 to be activated by a diverse range of ligands contrasts with the strict ligand requirements of CXCR4. In principle, ligands could promote activation by providing more energetically favorable binding interactions with the receptor in the active R* or R'' conformations relative to the inactive R or R' conformations. Alternatively, ligands could destabilize the inactive R or R' conformations of the receptor, which would also shift the receptor population to the active conformation(s). Our results do not distinguish between these two possibilities. Regardless, the similar free energies of different receptor conformations in ACKR3 and low energy barriers between them (Fig. 6) implies that relatively little energy is required to switch the receptor from one state to another. Moreover, activation can be achieved by populating either the R* or R'' state. Thus, independent of specific binding poses and receptor interactions, a variety of different ligands may be able to easily tip the balance between different receptor conformations. Such a mechanism would thereby promote activation and explain the observed ligand promiscuity of ACKR3. In contrast, the more rugged energy landscape of CXCR4, with larger free energy differences between R, R', and R* conformations and higher energy barriers between them, would require more ligand binding energy to promote receptor activation to the R* state. Additionally, not only is populating the R* conformation required for activation, it is not sufficient as illustrated for CXCL12P2G in Fig. 4B; instead specific interactions between the ligand and receptor are required, as borne out by mutagenesis studies (Fig. S10) (21, 23).

Another unusual aspect of ACKR3 behavior is the failure to activate G proteins. Why ACKR3 does not couple to G proteins, at least in most cells, is unclear and not readily explained by differences in primary sequence, since insertion of a DRY box motif and substitution of all ICLs from a canonical GPCR failed to confer G protein activity (16). It is possible that the active receptor conformation clashes sterically with the G protein as suggested by docking of G proteins to structures of ACKR3 (16). Alternatively, the receptor dynamics and conformational transitions revealed here may prevent formation of productive contacts between ACKR3 and G protein that are required for coupling, even though G proteins appear to constitutively associate with the receptor (13, 16, 67). Lack of a well-organized intracellular pocket due to frequent conformational transitions may also explain why the fingerloop of arrestin is not observed to interact with the pocket, in contrast with other GPCRs (68, 69), but instead inserts into membranes/micelles adjacent to the receptor (59). Nevertheless, arrestins are still recruited to CXCL12-stimulated ACKR3 due to GRK phosphorylation of the receptor C-terminal tail (15, 70). Since GRKs also interact with the cytoplasmic pocket to facilitate phosphorylation, it remains to be determined how dynamics might decouple G protein activation and arrestin binding to the receptor cytoplasmic pocket (59) while supporting pocket-mediated GRK activity. However, given the fleeting interaction between GRKs and GPCRs (71), rapid state sampling by ACKR3 may not necessarily be detrimental to GRK engagement and phosphorylation. Furthermore, conformational intermediates in addition to the fully active receptor have been shown to be targets for GRK phosphorylation, such as the early photoactivated rhodopsin metarhodopsin I (72). A more constrained system may be necessary to promote productive

interactions between ACKR3 and G proteins. Along these lines, a local increase of membrane pressure in certain cell environments could explain the apparent ability of ACKR3 to activate G proteins in astrocytes and glioma cells (12 [↗](#), 13 [↗](#)).

The pharmacological behavior of ACKR3 resembles the human cytomegalovirus chemokine receptor US28, which also recognizes diverse chemokines, constitutively internalizes, and displays multiple functional conformations (73 [↗](#)). Similar to ACKR3, US28 appears to be activated by distortion of the orthosteric binding pocket rather than by specific side chain contacts between receptor and ligand, which is supported by multiple active conformations observed for both apo-US28 and US28 with different agonists (73 [↗](#)–76 [↗](#)). Whether US28 also has a relatively flat energy landscape like ACKR3 remains to be seen. Such conformational dynamics and activation mechanisms may also be operative in other chemokine receptors that respond to multiple ligands and have considerable constitutive activity such as CCR1, CCR2 and CCR3 (61 [↗](#), 77 [↗](#)). Finally, the ability of ACKR3 to be easily activated by populating more than one conformational state may explain why antagonizing the receptor by targeting the orthosteric binding pocket has proven to be challenging; in contrast the specific requirements for CXCR4 agonism has permitted the development of many orthosteric antagonists but few agonists (78 [↗](#)). Drug discovery efforts aimed at inhibiting ACKR3 may therefore require allosteric strategies.

Materials and Methods

Unless otherwise stated all chemicals and reagents were purchased from SigmaAldrich or Fisher Scientific. Methoxy e-Coelenterazine (Prolume Purple) was purchased from Nanolight Technologies (Prolume LTD).

Cloning

ACKR3 (residues 2-362) preceded by an N-terminal HA signal sequence and followed by C-terminal 10His and FLAG tags was cloned into the pFasBac vector for purification from *Sf9* cells. CXCR4 (residues 2-352) with an N-terminal FLAG tag and C-terminal 10His was inserted into pFasBac for purification. For cell-based assays, ACKR3 (residues 2-362) or CXCR4 (residues 2-352) were inserted into pcDNA3.1 expression vector with an N-terminal FLAG tag and followed with a C-terminal *Renilla* luciferase II (ACKR3_rLucII and CXCR4_rLucII). Site-directed mutagenesis was performed by overlap extension and confirmed by Sanger sequencing.

Arrestin recruitment by BRET

Arrestin recruitment to ACKR3 and CXCR4 was detected using a BRET2 assay as previously described (15 [↗](#), 54 [↗](#)). Briefly, HEK293T cells (ATCC) were plated at 750k/well in a 6-well dish in Dulbecco's modified eagle media (DMEM) with 10% fetal bovine serum (FBS) and transfected 24 hrs later with 50 ng receptor_rLucII DNA, 1 μ g GFP10- β -arrestin2 (a kind gift from N. Heveker, Université de Montréal, Canada), and 1.4 μ g empty pcDNA3.1 vector using TransIT-LT1 transfection system (MirusBio) and expressed for 40 hrs. The cells were then washed with PBS (137 mM NaCl, 2.7 mM KCl, 10 mM Na₂HPO₄, 1.8 mM KH₂PO₄, pH 7.4) and mechanically lifted in Tyrode's buffer (25 mM HEPES, 140 mM NaCl, 1 mM CaCl₂, 2.7 mM KCl, 12 mM NaHCO₃, 5.6 mM Glucose, 0.5 mM MgCl₂, 0.37 mM NaH₂PO₄, pH 7.5). 100k cells were plated per 96-well white BRET plate (BD Fisher) and reattached for 45 min at 37 °C. GFP expression was checked using a SpectraMax M5 plate fluorometer (Molecular Devices) with 485 nm excitation, 538 nm emission, and 530 nm cutoff. 5 μ M Prolume Purple substrate was subsequently added and total luminescence detected using a TECAN Spark Luminometer (TECAN Life Sciences) at 37 °C. CXCL12 was then added to each well at the indicated final concentrations and BRET was read using default BRET2 settings (blue emission 360-440 nm, red emission 505-575 nm) and an integration time of

0.5 sec. Experiments were baseline matched and normalized to the Emax of WT receptor. The reported data is the average of three independent experiments performed in duplicate. Points were fit to a sigmoidal dose-response model using SigmaPlot 11.0 (Systat Software, Inc).

Receptor purification, labeling, and nanodisc reconstitution

M159^{4.40}C/Q245^{6.28}C ACKR3 (WT and Y257^{6.40}L) and L150^{4.40}C/Q233^{6.29}C CXCR4 were purified from *Sf9* cells (Expression Systems) as previously described (16). Briefly, *Sf9* cells were infected with baculovirus (prepared using Bac-to-Bac Baculovirus Expression System, Invitrogen) containing either the mutant ACKR3 or CXCR4. Cells were harvested after 48 hrs and membranes dounce homogenized in hypotonic buffer (10 mM HEPES pH 7.5, 10 mM MgCl₂, 20 mM KCl) followed three more times with hypotonic buffer with 1 M NaCl. The membranes were spun down at 50k x g for 30 min and resuspended between each round of douncing. After the final round, membranes were incubated with 100 μM CCX662 (Chemocentryx Inc.) for ACKR3 or 100 μM IT1t for CXCR4 and solubilized in 50 mM HEPES pH 7.5, 400 mM NaCl, 0.75/0.15% dodecyl maltoside/cholesteryl hemisuccinate (DDM/CHS) with a protease inhibitor tablet (Roche) for 4 hrs. Insoluble material was then removed by centrifugation at 50k x g for 30 min and Talon resin (Clontech) with 20 mM imidazole overnight binding at 4 °C. The resin was then transferred to a column and washed with WB1 (50 mM HEPES pH 7.5, 400 mM NaCl, 0.1/0.02% DDM/CHS, 10% glycerol, 20 mM imidazole) followed by WB2 (WB1 with 0.025/0.005% DDM/CHS) and finally eluted with WB2 with 250 mM imidazole. The imidazole was removed by desalting column (PD MiniTrap G-25, GE Healthcare). Final protein concentration was determined by A280 using an extinction coefficient of 75000 M⁻¹cm⁻¹ (ACKR3) and 58850 M⁻¹cm⁻¹ (CXCR4). Samples were snap frozen in liquid nitrogen, and stored at -80 °C until use.

When ready to prepare samples for smFRET measurements, two nanomoles of receptor was thawed and incubated with fourteen nanomoles of Alexa Fluor 555 (A555) and Cy5 overnight at 4 °C with rotation. The next morning, free label was removed by dilution using a 100k Da cut-off spin concentrator (Amicon) and the sample concentrated to ~100 μl. Label incorporation was evaluated by measuring the absorbance at A280 ($\epsilon_{280} = 75000 \text{ M}^{-1}\text{cm}^{-1}$ for ACKR3 and 58850 M⁻¹cm⁻¹ for CXCR4), A555 ($\epsilon_{555} = 150000 \text{ M}^{-1}\text{cm}^{-1}$), and A645 ($\epsilon_{645} = 250000 \text{ M}^{-1}\text{cm}^{-1}$) to detect the concentrations of the labeled receptor, A555, and Cy5, respectively. The contribution of the fluorophores to A280 was removed before determining labeled receptor concentration. The entire sample was used for nanodisc reconstitution.

Labeled receptors were reconstituted into biotinylated MSP1E3D1 nanodiscs as previously described (16). Briefly, 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phosphocholine (POPC, Avanti) and 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phospho-(1'-rac-glycerol) (POPG, Avanti) were prepared in a 3:2 POPC:POPG ratio and solubilized in ND buffer (25 mM HEPES pH 7.5, 150 mM NaCl, 180 mM cholate). MSP1E3D1 was expressed and purified as previously described (79) and biotinylated with EZ-Link NHS-polyethylene glycol 4 (PEG4)-Biotin (Thermo Fisher) per manufacturer instructions. The receptors, MSP1E3D1, and lipids were combined at a molar ratio of 0.1:1:110 for ACKR3:MSP:lipids respectively. Additional ND buffer was added to keep the final cholate concentration >20 mM. After 30 min at 4 °C, 200 mg of Biobeads (Bio-Rad) were added and incubated for 3-6 hrs. The sample was then loaded on a Superdex 200 10/300 GL column equilibrated with 25 mM HEPES pH 7.5, 150 mM NaCl and fractions containing nanodisc complexes were combined. 200 μl Talon resin was added with 20 mM imidazole (final concentration) and the samples were incubated for 16 hrs at 4 °C. The resin was then transferred to a Micro Bio-Spin Column (Bio-Rad) and washed with 25 mM HEPES pH 7.5, 150 mM NaCl, 20 mM imidazole and eluted with 25 mM HEPES pH 7.5, 150 mM NaCl, 250 mM imidazole. Imidazole was removed by buffer-exchange using 100k Da spin concentrators and 25 mM HEPES pH 7.5, 150 mM NaCl. Sample were concentrated to ~1 μM and stored at 4 °C until use.

CXCL12 purification from *E. Coli*

CXCL12 was expressed and purified as previously described (15, 16). Briefly, the chemokines were expressed by IPTG induction in BL21(DE3)pLysS cells. The cells were collected by centrifugation, resuspended in 50 mM tris pH 7.5, 150 mM NaCl and lysed by sonication. Inclusion bodies were then collected by centrifugation, resuspended in equilibration buffer (50 mM tris, 6 M guanidine-HCl pH 8.0), sonicated to release the chemokines and the samples centrifuged again to pellet insoluble material. The supernatant was then passed over a Ni-nitrilotriacetic acid (NTA) column equilibrated with equilibration buffer to bind the His-tagged chemokines. The column was washed with wash buffer (50 mM MES pH 6.0, 6 M guanidine-HCl) and eluted with 50 mM acetate pH 4.0, 6 M guanidine-HCl. The chemokine-containing elutions were pooled and dithiothreitol (DTT) added to a final concentration of 4 mM. After incubating 10 min, the solution was added dropwise into refolding buffer (50 mM tris pH 7.5, 500 mM arginine-HCl, 1 mM EDTA, 1 mM oxidized glutathione) and incubated at room temperature for 4 hrs before dialyzing against 20 mM tris pH 8.0, 50 mM NaCl. To remove the N-terminal purification tag, enterokinase was added and the sample incubated at 37 °C for 5 days. Uncleaved chemokine and free tags were removed by reverse Ni-NTA and eluted with wash buffer. Finally, the sample was purified on a reverse-phase C18 column equilibrated with 75% buffer A (0.1% trifluoroacetic acid (TFA)) and 25% buffer B (0.1% TFA, 90% acetonitrile) and eluted by a linear gradient of buffer B. The pure protein was lyophilized and stored at -80 °C until use.

smFRET microscopy

smFRET experiments were performed on a custom built prism-based TIRF microscope as previously described (80). Briefly, a flow cell was assembled on a quartz slide passivated with polyethylene glycol (PEG) and a small fraction of biotinylated PEG, after which neutravidin was introduced (81). Labeled ACKR3 or CXCR4 in biotinylated nanodiscs were diluted into trolox buffer (25 mM HEPES pH 7.5, 150 mM NaCl, 1 mM propyl gallate, 5 mM trolox), flowed into the sample chamber, and incubated for 5 min at room temperature. The samples were then washed twice with imaging buffer (trolox buffer with 2 mM protocatechuic acid, 50 nM protocatechuate-3,4-dioxygenase). Movies were collected with 100 ms integration time using a custom single-molecule data acquisition program to control the CCD camera (Andor). Single-molecule donor and acceptor emission traces were extracted from the recordings using custom IDL (Interactive Data Language) scripts. The software packages used to control the CCD camera and extract time trajectories were provided by Dr. Taekjip Ha. In all cases, five initial apo-receptor movies were recorded at different locations on the slide and then ligand was flowed into the cell by two washes with chemokine or small molecule in imaging buffer at final concentrations of 500 nM for chemokines (CXCL12WT, CXCL12P2G, CXCL12LRHQ, CXCL11) and 1 μM for the small molecules (IT1t, VUF16840, VUF15485). Ten more movies were collected for each condition at different locations on the slide. The data presented are a composite of at least three individual slides and treatments.

FRET trajectories were generated and analyzed using custom software written in-house (<https://github.com/rpauzek/smtirf>). Donor and acceptor traces for each molecule were corrected for donor bleed through and background signal and FRET efficiencies were calculated as $E = IA/(IA+ID)$, where E is the apparent FRET efficiency at each time point and ID and IA are the corresponding donor and acceptor fluorophore intensities, respectively. Traces were screened manually for single donor and acceptor bleach steps and anti-correlated behavior between fluorophores to confirm the presence of single receptors within the identified particles and single donor and acceptor labeling. All traces for a particular protein/ligand combination were analyzed globally by a single Hidden Markov Model assuming two, three, four, or five states and shared variance as previously described (80). Briefly, each model was trained on all selected trajectories for a given sample simultaneously using an expectation-maximization method (82). Once a model was trained, the Viterbi algorithm (82) was used to determine the most likely

hidden path for each trajectory. This labeled state path was then used to aggregate all data points belonging to a particular state in order to compile composite histograms of FRET efficiency, using a Kernel Density estimation algorithm (Python package Scikit-learn, version 1.2.2) with a Gaussian kernel and a bandwidth of 0.04. The relative populations of distinct FRET states were directly obtained during compilation of the corresponding histograms. The resulting FRET efficiency histograms were fit with a Gaussian Mixture Model (Scikit-learn) and the corresponding Bayesian Information Criterion (BIC) was calculated as described (83 [↗](#)). Transition density probability plots revealing the connectivity among individual FRET states were calculated as described (84 [↗](#)).

Data availability

All relevant data supporting this study are included in the Article or Supplementary files and are available from the authors upon request.

Acknowledgements

We gratefully acknowledge the initial work on this project undertaken by Chunxia Zhao and Rajan Lamichhane. Additionally, we thank Handel lab members Cheyanne Shinn, Catherina Salanga, and Nicholas Chimileski for providing the chemokines CXCL11 and CXCL12P2G, respectively, R. Leurs (Vrije Universiteit Amsterdam) for the small molecules VUF16840 and VUF15485, and N. Heveker (Université de Montréal) for the β -arrestin2_GFP10 plasmid. This work was supported by R01 GM133157 (D.P.M./T.M.H.), R01 CA254402 (T.M.H.), R01 AI161880 (T.M.H.), F32 GM137505 (C.T.S.), Robertson Foundation/Cancer Research Institute Irvington Postdoctoral Fellowship (M.G.), F32 GM115017 and T32 AI007354 (R.F.P.).

Author Contributions

Conception and design of the research: C.T.S., T.M.H., and D.P.M. Methodology: C.T.S., R.F.P., and M.G.

Software: R.F.P.

Validation: C.T.S. and R.F.P. Formal Analysis: C.T.S. and R.F.P. Investigation: C.T.S.

Writing-Original Draft Preparation: C.T.S., T.M.H., and D.P.M.

Writing-Reviewing & Editing: C.T.S., R.F.P., M.G., T.M.H., and D.P.M. Visualization: C.T.S. and R.F.P.

Supervision: T.M.H. and D.P.M. Funding Acquisition: T.M.H. and D.P.M.

Competing Interest Statement

T.M.H. is a cofounder of Lassogen Inc. and serves on the Scientific Advisory Boards of Abilita Bio, and Abalone Bio. The terms of these arrangements have been reviewed and approved by the University of California, San Diego in accordance with its conflict of interest policies. The other authors declare that they have no competing interests.

References

1. Chatterjee S., Behnam Azad B., Nimmagadda S. (2014) **The intricate role of CXCR4 in cancer** *Adv Cancer Res* **124**:31–82
2. Balkwill F. (2004) **The significance of cancer cell expression of the chemokine receptor CXCR4** *Semin Cancer Biol* **14**:171–179
3. Domanska U. M., et al. (2013) **A review on CXCR4/CXCL12 axis in oncology: no place to hide** *Eur J Cancer* **49**:219–230
4. Kawaguchi N., Zhang T. T., Nakanishi T. (2019) **Involvement of CXCR4 in Normal and Abnormal Development** *Cells* **8**
5. Kufareva I., Gustavsson M., Zheng Y., Stephens B. S., Handel T. M. (2017) **What Do Structures Tell Us About Chemokine Receptor Function and Antagonism?** *Annu Rev Biophys* **46**:175–198
6. Griffith J. W., Sokol C. L., Luster A. D. (2014) **Chemokines and chemokine receptors: positioning cells for host defense and immunity** *Annu Rev Immunol* **32**:659–702
7. Nibbs R. J., Graham G. J. (2013) **Immune regulation by atypical chemokine receptors** *Nat Rev Immunol* **13**:815–829
8. Vacchini A., Locati M., Borroni E. M. (2016) **Overview and potential unifying themes of the atypical chemokine receptor family** *J Leukoc Biol* **99**:883–892
9. Saaber F., et al. (2019) **ACKR3 Regulation of Neuronal Migration Requires ACKR3 Phosphorylation, but Not beta-Arrestin** *Cell Rep* **26**:1473–1488
10. Lau S., et al. (2020) **A negative-feedback loop maintains optimal chemokine concentrations for directional cell migration** *Nat Cell Biol* **22**:266–273
11. Wong M., et al. (2020) **Dynamic Buffering of Extracellular Chemokine by a Dedicated Scavenger Pathway Enables Robust Adaptation during Directed Tissue Migration** *Dev Cell* **52**:492–508
12. Odemis V., et al. (2012) **The presumed atypical chemokine receptor CXCR7 signals through G(i/o) proteins in primary rodent astrocytes and human glioma cells** *Glia* **60**:372–381
13. Fumagalli A., et al. (2020) **The atypical chemokine receptor 3 interacts with Connexin 43 inhibiting astrocytic gap junctional intercellular communication** *Nat Commun* **11**
14. Rajagopal S., et al. (2010) **Beta-arrestin- but not G protein-mediated signaling by the "decoy" receptor CXCR7** *Proc Natl Acad Sci U S A* **107**:628–632
15. Schafer C. T., Chen Q., Tesmer J. J. G., Handel T. M. (2023) **Atypical Chemokine Receptor 3 'Senses' CXC Chemokine Receptor 4 Activation Through GPCR Kinase Phosphorylation** *Mol Pharmacol* <https://doi.org/10.1124/molpharm.123.000710>

16. Yen Y. C., et al. (2022) **Structures of atypical chemokine receptor 3 reveal the basis for its promiscuity and signaling bias** *Sci Adv* **8**
17. Hopkins B. E., et al. (2022) **Effects of Small Molecule Ligands on ACKR3 Receptors** *Mol Pharmacol* **102**:128–138
18. Naumann U., et al. (2010) **CXCR7 functions as a scavenger for CXCL12 and CXCL11** *PLoS One* **5**
19. Hanes M. S., et al. (2015) **Dual targeting of the chemokine receptors CXCR4 and ACKR3 with novel engineered chemokines** *J Biol Chem* **290**:22385–22397
20. Jaracz-Ros A., et al. (2020) **Differential activity and selectivity of N-terminal modified CXCL12 chemokines at the CXCR4 and ACKR3 receptors** *J Leukoc Biol* <https://doi.org/10.1002/JLB.2MA0320-383RR>
21. Ngo T., et al. (2020) **Crosslinking-guided geometry of a complete CXC receptor- chemokine complex and the basis of chemokine subfamily selectivity** *PLoS Biol* **18**
22. Wescott M. P., et al. (2016) **Signal transmission through the CXC chemokine receptor 4 (CXCR4) transmembrane helices** *Proc Natl Acad Sci U S A* **113**:9928–9933
23. Stephens B. S., Ngo T., Kufareva I., Handel T. M. (2020) **Functional anatomy of the full- length CXCR4-CXCL12 complex systematically dissected by quantitative model- guided mutagenesis** *Sci Signal* **13**
24. Saotome K., et al. (2024) **Structural insights into CXCR4 modulation and oligomerization** *bioRxiv* <https://doi.org/10.1101/2024.02.09.579708>
25. Liu Y., et al. (2023) **Cryo-EM structure of monomeric CXCL12-bound CXCR4 in active state** *bioRxiv*
26. Burns J. M., et al. (2006) **A novel chemokine receptor for SDF-1 and I-TAC involved in cell survival, cell adhesion, and tumor development** *J Exp Med* **203**:2201–2213
27. Klein K. R., et al. (2014) **Decoy receptor CXCR7 modulates adrenomedullin-mediated cardiac and lymphatic vascular development** *Dev Cell* **30**:528–540
28. Ikeda Y., Kumagai H., Skach A., Sato M., Yanagisawa M. (2013) **Modulation of circadian glucocorticoid oscillation via adrenal opioid-CXCR7 signaling alters emotional behavior** *Cell* **155**:1323–1336
29. Meyrath M., et al. (2020) **The atypical chemokine receptor ACKR3/CXCR7 is a broad-spectrum scavenger for opioid peptides** *Nat Commun* **11**
30. Liu J. J., Horst R., Katritch V., Stevens R. C., Wuthrich K. (2012) **Biased signaling pathways in beta2-adrenergic receptor characterized by 19F-NMR** *Science* **335**:1106–1110
31. Wingler L. M., et al. (2019) **Angiotensin Analogs with Divergent Bias Stabilize Distinct Receptor Conformations** *Cell* **176**:468–478
32. Fay J. F., Farrens D. L. (2015) **Structural dynamics and energetics underlying allosteric inactivation of the cannabinoid receptor CB1** *Proc Natl Acad Sci U S A* **112**:8469–8474

33. Yao X., et al. (2006) **Coupling ligand structure to specific conformational switches in the beta2-adrenoceptor** *Nat Chem Biol* **2**:417–422
34. Elgeti M., Hubbell W. L. (2021) **DEER Analysis of GPCR Conformational Heterogeneity** *Biomolecules* **11**
35. Wingler L. M., et al. (2020) **Angiotensin and biased analogs induce structurally distinct active conformations within a GPCR** *Science* **367**:888–892
36. Ray A. P., Thakur N., Pour N. G., Eddy M. T. (2023) **Dual mechanisms of cholesterol- GPCR interactions that depend on membrane phospholipid composition** *Structure* **31**:836–847
37. Schafer C. T., Fay J. F., Janz J. M., Farrens D. L. (2016) **Decay of an active GPCR: Conformational dynamics govern agonist rebinding and persistence of an active, yet empty, receptor state** *Proc Natl Acad Sci U S A* **113**:11961–11966
38. Gregorio G. G., et al. (2017) **Single-molecule analysis of ligand efficacy in beta(2)AR-G-protein activation** *Nature* **547**:68–73
39. Krishna Kumar K., et al. (2023) **Negative allosteric modulation of the glucagon receptor by RAMP2** *Cell* **186**:1465–1477
40. Fernandes D. D., et al. (2021) **Ligand modulation of the conformational dynamics of the A(2A) adenosine receptor revealed by single-molecule fluorescence** *Sci Rep* **11**
41. Maslov I., et al. (2023) **Sub-millisecond conformational dynamics of the A(2A) adenosine receptor revealed by single-molecule FRET** *Commun Biol* **6**
42. Zhao J., et al. (2024) **Ligand efficacy modulates conformational dynamics of the micro-opioid receptor** *Nature* <https://doi.org/10.1038/s41586-024-07295-2>
43. Denisov I. G., Sligar S. G. (2016) **Nanodiscs for structural and functional studies of membrane proteins** *Nat Struct Mol Biol* **23**:481–486
44. Denisov I. G., Sligar S. G. (2017) **Nanodiscs in Membrane Biochemistry and Biophysics** *Chem Rev* **117**:4669–4713
45. Eberle S. A., Gustavsson M. (2024) **Bilayer lipids modulate ligand binding to atypical chemokine receptor 3** *Structure* <https://doi.org/10.1016/j.str.2024.04.018>
46. Tsukamoto H., Sinha A., DeWitt M., Farrens D. L. (2010) **Monomeric rhodopsin is the minimal functional unit required for arrestin binding** *J Mol Biol* **399**:501–511
47. Eberle S. A., Gustavsson M. (2022) **A Scintillation Proximity Assay for Real-Time Kinetic Analysis of Chemokine-Chemokine Receptor Interactions** *Cells* **11**
48. Rasmussen S. G., et al. (2011) **Crystal structure of the beta2 adrenergic receptor-Gs protein complex** *Nature* **477**:549–555
49. Palczewski K., et al. (2000) **Crystal structure of rhodopsin: A G protein-coupled receptor** *Science* **289**:739–745
50. Rasmussen S. G., et al. (2007) **Crystal structure of the human beta2 adrenergic G- protein-coupled receptor** *Nature* **450**:383–387

51. Farrens D. L., Altenbach C., Yang K., Hubbell W. L., Khorana H. G. (1996) **Requirement of rigid-body motion of transmembrane helices for light activation of rhodopsin** *Science* **274**:768–770
52. Kleist A. B., et al. (2022) **Conformational selection guides beta-arrestin recruitment at a biased G protein-coupled receptor** *Science* **377**:222–228
53. Otun O., et al. (2023) **Conformational dynamics underlying Atypical Chemokine Receptor 3 activation** *BioRxiv*
54. Gustavsson M., Dyer D. P., Zhao C., Handel T. M. (2019) **Kinetics of CXCL12 binding to atypical chemokine receptor 3 reveal a role for the receptor N terminus in chemokine binding** *Sci Signal* **12**
55. Han M., Smith S. O., Sakmar T. P. (1998) **Constitutive activation of opsin by mutation of methionine 257 on transmembrane helix 6** *Biochemistry* **37**:8253–8261
56. Han X., Tachado S. D., Koziel H., Boisvert W. A. (2012) **Leu128(3.43) (I128) and Val247(6.40) (V247) of CXCR1 are critical amino acid residues for g protein coupling and receptor activation** *PLoS One* **7**
57. Cui M., et al. (2022) **Crystal structure of a constitutive active mutant of adenosine A(2A) receptor** *IUCrj* **9**:333–341
58. Tsukamoto H., Farrens D. L. (2013) **A constitutively activating mutation alters the dynamics of a key conformational change in a ligand-free GPCR** *J Biol Chem* <https://doi.org/10.1074/jbc.M113.472464>
59. Chen Q., et al. (2023) **ACKR3-arrestin2/3 complexes reveal molecular consequences of GRK-dependent barcoding** *bioRxiv* <https://doi.org/10.1101/2023.07.18.549504>
60. Deupi X., et al. (2012) **Stabilized G protein binding site in the structure of constitutively active metarhodopsin-II** *Proc Natl Acad Sci U S A* **109**:119–124
61. Shao Z., et al. (2022) **Molecular insights into ligand recognition and activation of chemokine receptors CCR2 and CCR3** *Cell Discov* **8**
62. Qin L., et al. (2015) **Crystal structure of the chemokine receptor CXCR4 in complex with a viral chemokine** *Science* **347**:1117–1122
63. Liu K., et al. (2020) **Structural basis of CXC chemokine receptor 2 activation and signalling** *Nature* **585**:135–140
64. Szpakowska M., et al. (2018) **Mutational analysis of the extracellular disulphide bridges of the atypical chemokine receptor ACKR3/CXCR7 uncovers multiple binding and activation modes for its chemokine and endogenous non-chemokine agonists** *Biochem Pharmacol* **153**:299–309
65. Ai L. S., Liao F. (2002) **Mutating the four extracellular cysteines in the chemokine receptor CCR6 reveals their differing roles in receptor trafficking, ligand binding, and signaling** *Biochemistry* **41**:8332–8341
66. Limatola C., et al. (2005) **Cysteine residues are critical for chemokine receptor CXCR2 functional properties** *Exp Cell Res* **307**:65–75

67. Levoye A., Balabanian K., Baleux F., Bachelier F., Lagane B. (2009) **CXCR7 heterodimerizes with CXCR4 and regulates CXCL12-mediated G protein signaling** *Blood* **113**:6085–6093
68. Staus D. P., et al. (2020) **Structure of the M2 muscarinic receptor-beta-arrestin complex in a lipid nanodisc** *Nature* **579**:297–302
69. Huang W., et al. (2020) **Structure of the neurotensin receptor 1 in complex with beta-arrestin 1** *Nature* **579**:303–308
70. Sarma P., et al. (2023) **Molecular insights into intrinsic transducer-coupling bias in the CXCR4-CXCR7 system** *Nat Commun* **14**
71. Pulvermuller A., Palczewski K., Hofmann K. P. (1993) **Interaction between photoactivated rhodopsin and its kinase: stability and kinetics of complex formation** *Biochemistry* **32**:14082–14088
72. Paulsen R., Bontrop J. (1983) **Activation of rhodopsin phosphorylation is triggered by the lumirhodopsin-metarhodopsin I transition** *Nature* **302**:417–419
73. De Groof T. W. M., et al. (2021) **Selective targeting of ligand-dependent and - independent signaling by GPCR conformation-specific anti-US28 intrabodies** *Nat Commun* **12**
74. Burg J. S., et al. (2015) **Structural basis for chemokine recognition and activation of a viral G protein-coupled receptor** *Science* **347**:1113–1117
75. Miles T. F., et al. (2018) **Viral GPCR US28 can signal in response to chemokine agonists of nearly unlimited structural degeneracy** *Elife* **7**
76. Tsutsumi N., et al. (2022) **Atypical structural snapshots of human cytomegalovirus GPCR interactions with host G proteins** *Sci Adv* **8**
77. Gilliland C. T., Salanga C. L., Kawamura T., Trejo J., Handel T. M. (2013) **The chemokine receptor CCR1 is constitutively active, which leads to G protein- independent, beta-arrestin-mediated internalization** *J Biol Chem* **288**:32194–32210
78. Lefrancois M., et al. (2011) **Agonists for the Chemokine Receptor CXCR4** *ACS Med Chem Lett* **2**:597–602
79. Ritchie T. K., et al. (2009) **Chapter 11 - Reconstitution of membrane proteins in phospholipid bilayer nanodiscs** *Methods Enzymol* **464**:211–231
80. Pauszek R. F. (2021) **3rd, R. Lamichhane, A. Rajkarnikar Singh, D. P. Millar, Single- molecule view of coordination in a multi-functional DNA polymerase** *Elife* **10**
81. Lamichhane R., Solem A., Black W., Rueda D. (2010) **Single-molecule FRET of protein- nucleic acid and protein-protein complexes: surface passivation and immobilization** *Methods* **52**:192–200
82. Rabiner L. R. (1989) **A Tutorial on Hidden Markov-Models and Selected Applications in Speech Recognition** *P IEEE* **77**:257–286
83. Schwarz G. (1978) **Estimating the Dimension of a Model** *Ann. Statist* **6**:461–464

84. McKinney S. A., Joo C., Ha T. (2006) **Analysis of single-molecule FRET trajectories using hidden Markov modeling** *Biophys J* **91**:1941–1951

Editors

Reviewing Editor

Volker Dötsch

Goethe University, Frankfurt am Main, Germany

Senior Editor

Volker Dötsch

Goethe University, Frankfurt am Main, Germany

Reviewer #1 (Public Review):

Summary:

This paper uses single-molecule FRET to investigate the molecular basis for the distinct activation mechanisms between 2 GPCR responding to the chemokine CXCL12 : CXCR4, that couples to G-proteins, and ACKR3, which is G-protein independent and displays a higher basal activity.

Strengths:

It nicely combines the state-of-the-art techniques used in the studies of the structural dynamics of GPCR. The receptors are produced from eukaryotic cells, mutated, and labeled with single molecule compatible fluorescent dyes. They are reconstituted in nanodiscs, which maintain an environment as close as possible to the cell membrane, and immobilized through the nanodisc MSP protein, to avoid perturbing the receptor's structural dynamics by the use of an antibody for example.

The smFRET data are analysed using the HHMI technique, and the number of states to be taken into account is evaluated using a Bayesian Information Criterion, which constitutes the state-of-the-art for this task.

The data show convincingly that the activation of the CXCR4 and ACKR3 by an agonist leads to a shift from an ensemble of high FRET states to an ensemble of lower FRET states, consistent with an increase in distance between the TM4 and TM6. The two receptors also appear to explore a different conformational space. A wider distribution of states is observed for ACKR3 as compared to CXCR4, and it shifts in the presence of agonists toward the active states, which correlates well with ACKR3's tendency to be constitutively active. This interpretation is confirmed by the use of the mutation of Y254 to leucine (the corresponding residue in CXCR4), which leads to a conformational distribution that resembles the one observed with CXCR4. It is correlated with a decrease in constitutive activity of ACKR3.

Weaknesses:

Although the data overall support the claims of the authors, there are however some details in the data analysis and interpretation that should be modified, clarified, or discussed in my opinion.

Concerning the amplitude of the changes in FRET efficiency: the authors do not provide any structural information on the amplitude of the FRET changes that are expected. To me, it looks like a FRET change from ~0.9 to ~0.1 is very important, for a distance change that is expected to be only a few angstroms concerning the movement of the TM6. Can the authors

give an explanation for that? How does this FRET change relate to those observed with other GPCRs modified at the same or equivalent positions on TM4 and TM6?

Concerning the intermediate states: the authors observe several intermediate states.

(1) First I am surprised, looking at the time traces, by the dwell times of the transitions between the states, which often last several seconds. Is such a long transition time compatible with what is known about the kinetic activation of these receptors?

(2) Second is it possible that these « intermediate » states correspond to differences in FRET efficiencies, that arise from different photophysical states of the dyes? Alexa555 and Cy5 are Cyanines, that are known to be very sensitive to their local environment. This could lead to different quantum yields and therefore different FRET efficiencies for a similar distance. In addition, the authors use statistical labeling of two cysteines, and have therefore in their experiment a mixture of receptors where the donor and acceptor are switched, and can therefore experience different environments. The authors do not speculate structurally on what these intermediate states could be, which is appreciated, but I think they should nevertheless discuss the potential issue of fluorophore photophysics effects.

(3) It would also have been nice to discuss whether these types of intermediate states have been observed in other studies by smFRET on GPCR labeled at similar positions.

On line 239: the authors talk about the $R \leftrightarrow R'$ transitions that are more probable. In fact it is more striking that the $R' \leftrightarrow R^*$ transition appears in the plot. This transition is a signature of the behaviour observed in the presence of an agonist, although IT1t is supposed to be an inverse agonist. This observation is consistent with the unexpected (for an inverse agonist) shift in the FRET histogram distribution. In fact, it appears that all CXCR4 antagonists or inverse agonists have a similar (although smaller) effect than the agonist. Is this related to the fact that these (antagonist or inverse agonist) ligands lead to a conformation that is similar to the agonists, but cannot interact with the G-protein ?? Maybe a very interesting experiment would be here to repeat these measurements in the presence of purified G-protein. G-protein has been shown to lead to a shift of the conformational space explored by GPCR toward the active state (using smFRET on class A and class C GPCR). It would be interesting to explore its role on CXCR4 in the presence of these various ligands. Although I am aware that this experiment might go beyond the scope of this study, I think this point should be discussed nevertheless.

The authors also mentioned in Figure 6 that the energetic landscape of the receptors is relatively flat ... I do not really agree with this statement. For me, a flat conformational landscape would be one where the receptors are able to switch very rapidly between the states (typically in the submillisecond timescale, which is the timescale of protein domain dynamics). Here, the authors observed that the transition between states is in the second timescale, which for me implies that the transition barrier between the states is relatively high to preclude the fast transitions.

<https://doi.org/10.7554/eLife.100098.1.sa2>

Reviewer #2 (Public Review):

Summary:

This manuscript uses single-molecule fluorescence resonance energy transfer (smFRET) to identify differences in the molecular mechanisms of CXCR4 and ACKR3, two 7-transmembrane receptors that both respond to the chemokine CXCL12 but otherwise have very different signaling profiles. CXCR4 is highly selective for CXCL12 and activates heterotrimeric G proteins. In contrast, ACKR3 is quite promiscuous and does not couple to G

proteins, but like most G protein-coupled receptors (GPCRs), it is phosphorylated by GPCR kinases and recruits arrestins. By monitoring FRET between two positions on the intracellular face of the receptor (which highlights the movement of transmembrane helix 6 [TM6], a key hallmark of GPCR activation), the authors show that CXCR4 remains mostly in an inactive-like state until CXCL12 binds and stabilizes a single active-like state. ACKR3 rapidly exchanges among four different conformations even in the absence of ligands, and agonists stabilize multiple activated states.

Strengths:

The core method employed in this paper, smFRET, can reveal dynamic aspects of these receptors (the breadth of conformations explored and the rate of exchange among them) that are not evident from static structures or many other biophysical methods. smFRET has not been broadly employed in studies of GPCRs. Therefore, this manuscript makes important conceptual advances in our understanding of how related GPCRs can vary in their conformational dynamics.

Weaknesses:

- (1) The cysteine mutations in ACKR3 required to site-specifically install fluorophores substantially increase its basal and ligand-induced activity. If, as the authors posit, basal activity correlates with conformational heterogeneity, the smFRET data could greatly overestimate the conformational heterogeneity of ACKR3.
- (2) The probes used cannot reveal conformational changes in other positions besides TM6. GPCRs are known to exhibit loose allosteric coupling, so the conformational distribution observed at TM6 may not fully reflect the global conformational distribution of receptors. This could mask important differences that determine the ability of intracellular transducers to couple to specific receptor conformations.
- (3) While it is clear that CXCR4 and ACKR3 have very different conformational dynamics, the data do not definitively show that this is the main or only mechanism that contributes to their functional differences. There is little discussion of alternative potential mechanisms.
- (4) The extent to which conformational heterogeneity is a characteristic feature of ACKRs that contributes to their promiscuity and arrestin bias is unclear. The key residue the authors find promotes ACKR3 conformational heterogeneity is not conserved in most other ACKRs, but alternative mechanisms could generate similar heterogeneity.
- (5) There are no data to confirm that the two receptors retain the same functional profiles observed in cell-based systems following in vitro manipulations (purification, labeling, nanodisc reconstitution).

<https://doi.org/10.7554/eLife.100098.1.sa1>

Reviewer #3 (Public Review):

Summary:

This is a well-designed and rigorous comparative study of the conformational dynamics of two chemokine receptors, the canonical CXCR4 and the atypical ACKR3, using single-molecule fluorescence spectroscopy. These receptors play a role in cell migration and may be relevant for developing drugs targeting tumor growth in cancers. The authors use single-molecule FRET to obtain distributions of a specific intermolecular distance that changes upon activation of the receptor and track differences between the two receptors in the apo state, and in response to ligands and mutations. The picture emerging is that more dynamic

conformations promote more basal activity and more promiscuous coupling of the receptor to effectors.

Strengths:

The study is well designed to test the main hypothesis, the sample preparation and the experiments conducted are sound and the data analysis is rigorous. The technique, smFRET, allows for the detection of several substates, even those that are rarely sampled, and it can provide a "connectivity map" by looking at the transition probabilities between states. The receptors are reconstituted in nanodiscs to create a native-like environment. The examples of raw donor/acceptor intensity traces and FRET traces look convincing and the data analysis is reliable to extract the sub-states of the ensemble. The role of specific residues in creating a more flat conformational landscape in ACKR3 (e.g., Y257 and the C34-C287 bridge) is well documented in the paper.

Weaknesses:

The kinetics side of the analysis is mentioned, but not described and discussed. I am not sure why since the data contains that information. For instance, it is not clear if greater conformational flexibility is accompanied by faster transitions between states or not.

The method to choose the number of states seems reasonable, but the "similarity" of states argument (Figures S4 and S6) is not that clear.

Also, the "dynamics" explanation offered for ACKR3's failure to couple and activate G proteins is not very convincing. In other studies, it was shown that activation of GPCRs by agonists leads to an increase in local dynamics around the TM6 labelling site, but that did not prevent G protein coupling and activation.

<https://doi.org/10.7554/eLife.100098.1.sa0>

Author response:

The authors intend to submit a revised manuscript that addresses the questions raised in the public reviews.

<https://doi.org/10.7554/eLife.100098.1.sa4>